Robust Multiwavelength All-Fiber Source of Polarization-Entangled Photons With Built-In Analyzer Alignment Signal

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(Invited Paper)

***Abstract*—We describe an all-fiber source of polarization-entangled photons, which is designed to be simple to manufacture and structurally robust. The source includes an alignment signal that allows for straightforward alignment of the measurement axes of the subsequent polarization analyzers. High-quality entangled light is generated and measured at multiple wavelength channels within the C-band.**

***Index Terms*—Polarization-entangled state, quantum communications, quantum optics.**

I. Introduction

QUANTUM communications promise new applications such as quantum cryptography and quantum games, which have properties that classical systems cannot fully emulate. A core component in many quantum processes is entanglement. Entangled photons are particularly useful since they can be distributed over long lengths of a low-loss fiber optical cable. In order to advance the field, it is important to have reliable and easy-to-use high-quality sources of entangled photons that are compatible with distribution via fiber optics.

Research on sources of entangled light has advanced rapidly. Initially, bulk $\chi^{(2)}$ crystals were the nonlinear medium of choice [1] for generating entangled-photon pairs via spontaneous parametric down-conversion. Current $\chi^{(2)}$ crystal-based sources offer excellent performance, including large two-photon interference (TPI) fringe visibility of $>99\%$ with high measured coincidence count rates extending into the tens of kilohertz range when silicon single-photon detectors (SPDs) can be used [2]. Unfortunately, coincidence count rates for entangled photons in the telecommunication bands are generally much lower due to the poor relative performance of SPDs that are sensitive in the 1310 or 1550 nm bands. The reduced performance stems from either lower detection efficiency, or lower maximum counting rates, or both.

The probability of generating a biphoton at the source during a measurement interval must be suitably low in order to avoid multiphoton effects, which degrade the TPI. Biphoton generation rates of $<3\%$ are required for $>95\%$ visibilities [3]. Recently, $\chi^{(2)}$ waveguide devices have produced two-photon visibilities in the telecommunication band near the maximum levels allowed for their respective biphoton generation rates [3]. Actual measured coincidence count rates are reduced by the square of the losses incurred between the generation and subsequent measurement of the state. Thus, coupling losses from the waveguide to fiber as well as waveguide propagation losses can, in practice, reduce the counting rate considerably.

Since the first observation of quantum correlations generated directly in optical fiber [4], research into fiber-based entangled sources has greatly progressed [5]–[9]. Such sources spontaneously generate photon pairs via four-wave mixing using the $\chi^{(3)}$ nonlinearity of fiber. Fiber-based designs have the advantage of being inherently compatible with fiber distribution systems and potentially simple to manufacture. They also have excellent modal purity, which could be important for cascaded quantum operations. Fiber-based nonlinearities, which create biphotons in the 1550 and/or 1310 nm bands, have the disadvantage that Raman scattering causes some noise photons that degrade the performance. Raman scattering can be mitigated by spacing the signal and idler wavelengths suitably close together where the Raman gain is small [10]. It can be further reduced by cooling the fiber and thereby suppressing the thermal phonon bath [6], [9]. The other technique to reduce Raman scattering in fibers is to space the signal and idler very far apart so that they are outside the Raman gain peak [8]. However, while increasing the signal/idler wavelength spacing has produced high-quality room-temperature sources, it does not allow for both the signal and idler to be in the telecommunication wavebands, as in that case, the pump-to-signal separation is not large enough to suppress Raman scattering adequately.

Telecommmunication-band signal and idler photons are useful for quantum networks, which may make use of a centrally located entangled-photon source (EPS) to distribute entanglement to multiple users [11]. $\chi^{(2)}$ crystals and photonic crystal fiber have been used to generate broadband entanglement that could generate multiple entangled pairs at different wavelength channels using appropriate dense wavelength-division multiplexing (DWDM) technology [2], [8], [12]. In this paper, we characterize the performance of three distinct channels of entangled-photon pairs in the 1550 nm band from a single solid-core fiber source, thus demonstrating that traditional fiber can also be used in such applications.

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Our focus is on a telecommunication-band fiber-based EPS design that uses all commercially made fiber-coupled components. Manufacturing such a source should be highly reliable, and consists primarily of splicing the necessary components together. It is expected that the system could be made both rugged and compact. Such characteristics have obvious practical benefits.

The source described herein creates polarization-entangled photons. Detecting polarization entanglement is typically a process of lower cost and smaller insertion loss than detecting time-bin entanglement due to the maturity of polarization-control technology. While for time-bin entanglement, one must stabilize local interferometers [9], with polarization entanglement, it is the polarization drift caused by the transmission fiber that must be addressed. This, in turn, suggests that a tool for properly aligning the polarization analyzers that are used to measure the entanglement downstream should be integrated into a practical polarization-entangled source.

A previous experiment that distributed polarization entanglement for the purposes of quantum cryptography aligned the polarization analyzers using only coincident count measurements [13]. This method is not ideal as it is hampered by the relatively long time required to obtain a meaningful measurement since coincidence count rates are much less than the corresponding singles count rates. Another quantum cryptographic experiment made use of an external laser to help simplify the alignment [14]. The external laser was calibrated with a removable calibration device to be copolarized with the pump. The polarization analyzers could then be aligned to the external laser.

We improve upon prior work by integrating an alignment signal directly into the source, thereby creating an alignment signal on demand. By employing a depolarized broadband alignment source, we eliminate the need for any adjustable components when switching between an alignment signal and an entangled output. The polarization analyzers are configured to minimize the alignment signal power in order to align them to observe a TPI fringe. By making use of a variable phase retarder in one of the analyzers, TPI fringes in different polarization bases can be recorded.

II. System Design

A. Background

Both a quality and a rate metric can be used to characterize an EPS, with both metrics being pump-power-dependent. Various quality metrics exist, including the TPI fringe, which is based on measuring the coincidence counts as a function of measurement angle, and those derived from tomographic measurements, such as fidelity and tangle [2]. In this paper, we use the TPI visibility value. Visibility above 71% signifies a quantum effect that is, in principle, enough to be useful for quantum experiments and applications such as the violation of Bell's inequality or quantum communication. The value of TPI visibility that is acceptable for a given experiment depends on many specific factors. Visibility values of ~98% or higher are usually considered excellent. However, it is very difficult to maintain such high visibilities when distributing entanglement due to the combination of the detector dark noise and exponential loss in fibers [11].

The two-photon generation rate per pump pulse of the quantum process generating the two-photon pairs is a useful metric because it is directly connected to the TPI performance limit caused by multipair emissions [3]. While the two-photon generation rate is a useful characterization metric, it does not necessarily allow for direct comparisons between measured two-photon rates from different sources since certain losses internal to the source, such as coupling losses from the nonlinear medium to fiber or filtering losses, are not included. Additionally, the actual measured coincidence value is also highly dependent on the characteristics of the SPDs used as well as other types of linear losses that may exist between the source and detectors. For losses that are common to both the signal and idler photons, the detected biphoton rate is proportional to the square of the losses between generation and measurement. Thus, a reduction in detection efficiency of a factor of 2 leads to a fourfold decrease in the detected biphoton rate. Usually, the number of coincidence counts per unit time is limited by the speed/recovery time of the detectors used. Therefore, the detected two-photon generation rate is also related to the speed of the SPDs. Thus, the measured coincidence count rate is very much related to the entire system and is not necessarily reflective of the source itself. We note that fiber-based sources are not sensitive to power-dependent effects such as two-photon absorption. Therefore, they can be scaled to very high repetition rates [14], which would be useful if high-speed emerging detector technologies mature [15].

Particularly, when working at the telecommunication bands, various nonidealities of the detection process often need to be considered. Since we are interested only in characterizing the source, we will subtract dark counts from our measurement. Dark counts can vary significantly depending on the detection method used [7], [15], [16]. Previous experimental results of fiber entangled sources using ultralow dark count superconducting detectors are consistent with those of the same sources using higher dark count APDs after dark count subtraction [7]. Using fast APDs with dark count subtraction is thus a convenient method of accurately characterizing the source.

Note that in some published works, "accidental" counts are subtracted. In some experiments, accidental counts are synonymous with dark counts; however, this is not always the case. For instance, accidental counts can include coincidences, which may occur from various phenomena including some that are inherent to the entanglement generation process such as Raman noise photons. We subtract only dark counts, which allows us to characterize the source performance as opposed to the detector performance.

To measure the TPI, two SPDs are used: one to detect the signal and the other to detect the idler. The SPDs are preceded by polarization controllers and analyzers, consisting of wave plates and polarizers. The basic measurements are the singles counts $S_{x,y}$ and the raw coincidence counts C. Here, x represents the signal (s) or idler (i) photon and y represents either a dark count (d, no illumination) or total count (t, illuminated) condition.

The number of true singles counts after subtracting dark counts is thus simply $S_{x,t} - S_{x,d}$. The dark counts are

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determined by blocking all light to the detectors. The number of coincidence counts calculated after subtracting dark counts is $C_{\rm calc} = C - C_{\rm dark}$, where $C_{\rm dark} = (S_{s,t}S_{i,d}/G) + (S_{i,t}S_{s,d}/G) - (S_{s,d}S_{i,d}/G)$, and G is the number of gates sent to the APD in the counting interval (number of optical pulses arriving at an active detector). Here, $C_{\rm dark}$ are the coincidence counts caused by dark counts, assuming that the dark counts on one detector are statistically independent of the total singles counts on the other detector. The dark counts from the two SPDs are verified to be statistically independent by observing that the number of measured coincidence counts C is equal to the calculated $C_{\rm dark}$ when no light is incident on the detectors. Similarly, when each of the two uncorrelated light sources (i.e., highly attenuated lasers) illuminate one of the two detectors, $C_{\rm calc}$ was observed to equal the expected coincidence count rate for independent Poisson counting events—$C_{\rm Exp} = ((S_{s,t} - S_{s,d})/G)\,((S_{i,t} - S_{i,d})/G)\,G$. In the case of performing measurements for entangled photons, the TPI visibility can be determined by $V_{\rm TPI} = (C_{\max} - C_{\min})/(C_{\max} + C_{\min})$, where $C_{\max}$ and $C_{\min}$ are, respectively, the peak and trough of the TPI fringe obtained by plotting $C_{\rm calc}$ versus polarizer angle of the polarization analyzer.

One can also obtain a very quick estimate of the TPI visibility without taking a full TPI curve. The calculation relies on the fact that all the correlated coincidence counts from four-wave mixing should be blocked at the minimum position of the TPI fringe since the polarization analyzers are oriented orthogonal to each other. Therefore, the number of coincidence counts at the minimum of the TPI fringe should closely match the calculated accidental coincidence count level. The calculated accidental coincidence count level due to illumination is $A_{\rm calc} = A_{\rm singles} - C_{\rm dark}$, where $A_{\rm singles} = S_{s,t}S_{i,t}/G$. It follows that the estimated TPI visibility is $V_{\rm est} = (C_{\max} - A_{\rm calc})/(C_{\max} + A_{\rm calc})$. Note that since the numbers of singles counts are generally large (much higher than the coincidence count levels), these derived estimates have small variances in comparison to the measured coincidence count data. The only measurement required to estimate the TPI visibility is at the TPI fringe maximum ($C_{\max}$), and the resulting estimate is insensitive to slight misalignments of the wave plates. Since the coincidence rate is largest at the maximum coincidence point, the TPI can be estimated quickly. This measurement is related to the coincidence-to-accidental measurement commonly used in the literature [7].

The utility of estimating the TPI visibility without performing a full scan lies in quickly estimating the source's performance at various system settings. As we will see, performance is a strong function of pump power with the two-photon generation rate increasing quadratically with pump power, while the TPI visibility generally decreases with pump power. A user can choose the desired pump-power level to balance these effects. Any change in the system, such as changes in the propagation or detection losses, will change the relationship between the performance and pump power. The estimated TPI visibility allows one to quickly estimate the optimum pump value. We find that the resulting estimates are generally quite accurate, with the accuracy becoming less reliable for very low pump powers where the coincidence count rate is small (less than five counts per million gates). In general, if the full TPI fringe does not match well to the estimated TPI, then there is likely a misalignment in the source or measurement apparatus that would need to be addressed, or the system is experiencing excessive pump leakage into the signal or idler bands.

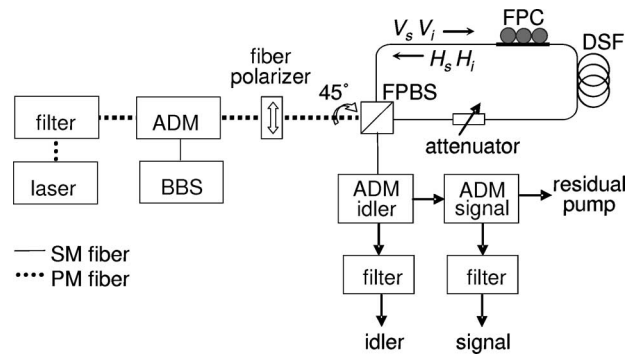

Fig. 1. Schematic diagram of the all-fiber polarization EPS. ADM is an add–drop multiplexer and BBS is a broadband optical source used for analyzer alignment purposes only.

B. Apparatus Construction

Fig. 1 shows a schematic of the EPS. It is a Sagnac loop architecture similar to [6], but without any pump-polarization control. The polarization of the pump is instead fixed by connecting the pump laser to the Sagnac loop using all polarization-maintaining (PM) fiber. The desired 50/50 split of the pump photons into the clockwise and counterclockwise propagating directions is obtained by orienting the PM fiber at the input to the fiber coupled polarization beam splitter (FPBS) such that the PM axis is at 45°. This fixed coupling method is very convenient as it requires no user alignment. However, with such a configuration, it would ordinarily be difficult to align the polarization analyzers subsequently used to measure the generated signal and idler photon pairs since the pairs are depolarized. It is possible to use coincidence counts for aligning the analyzers; however, this is difficult because coincidence count rates are low.

This issue is resolved by the inclusion of a broadband depolarized alignment light source. We use amplified spontaneous emission of an erbium-doped fiber amplifier as the broadband source, although inexpensive sources such as LEDs could also be used. The broadband source is combined with the pump laser using an add–drop multiplexer, and then copolarized with the pump laser using an inline polarizer. The alignment source has spectral components in both the signal and idler bands. After the alignment source propagates through the system and is filtered by the signal and idler filters, it has the same spectral profile as the entangled light and a polarization that is a 50/50 combination of the horizontal (H) and vertical (V) polarizations defined by the Sagnac loop FPBS. Thus, the polarized light has a fixed relation with respect to the orientation of the entangled light and can be used to aid in alignment. Additionally, there is no manual optical path manipulation between alignment and entanglement modes allowing for the system to be switched from one to the other simply by turning ON and OFF the alignment source.

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The pump laser is a 1554.1-nm, 50-MHz fiber-mode-locked laser from PolarOnyx, Inc. Its emission band is defined by an internally mounted WDM filter with $\sim$1 nm bandwidth. Additional pump filters are used to prevent any stray pump light from entering the signal/idler bands. The filtered pump light is passed through a PM fiber polarizer to clean up its polarization. The WDM filters and the fiber polarizer are stock components from AC Photonics, Inc.

Although coupled to a PM fiber, the polarization directly out of the laser drifts slightly over time; therefore, we do observe some long-term drift of the pump power after the polarizer. The change in pump power is exacerbated by the slight misalignments in the PM fiber splices and PM components before the fiber polarizer. These multiple segments of slightly misaligned PM sections act as a polarization-mode dispersion emulator, and as the ambient temperature drifts, the amount of instantaneous PMD changes as well, causing the power at the output of the polarizer to wander. The phenomenon can be seen in some of the results in this paper, but was of a manageable degree. Better alignment of the laser to the PM fiber should reduce the drift. Alternatively, the PM fiber–pump system could be replaced with all single-mode components followed by an electronic polarization controller to set the pump polarization at the polarizer more precisely.

After the polarizer, the pump light enters a FPBS with its input PM fiber rotated by 45°. This causes the pump light to be split evenly into clockwise and counterclockwise propagating waves. The actual measured splitting ratio of the cube is 51/49 due to manufacturing tolerances. All the fiber inside the loop is standard single-mode fiber.

A fiber polarization controller (FPC) inside the Sagnac loop is used to control the exit path of the photons generated inside the loop. Since the generated photons propagate in the same direction as the pump, this task is accomplished by adjusting the FPC for maximum residual pump power transmitted through the Sagnac Loop. Once set, the FPC is stable for many hours, although, clearly, an automated feedback mechanism could be used to periodically optimize the setting.

Although the FPC has a low insertion loss of $<$1 dB, an attenuator with an equal amount of loss as the FPC is added at the other end of the nonlinear dispersion shifted fiber (DSF) to balance the loss in the loop. This equalizes the nonlinear phase shift experienced by the pump in both directions, thus ensuring that the relative phase between the counterpropagating photons exiting the FPBS is not pump-power-dependent [8]. A nonlinear change in the relative phase would reduce the quality of basis alignment since the alignment signal has a low peak power, and thus, would not experience the same nonlinear phase shifts. Another reason for balancing the loss is to equalize the amount of photons generated in both directions in order for them to be maximally entangled at the output. The DSF is a 300-m-long fiber with a zero-dispersion wavelength near the pump wavelength.

The clockwise-propagating component of the pump produces vertically polarized photon pairs of the state $|V\rangle_s |V\rangle_i$, and the counterclockwise component produces $|H\rangle_s |H\rangle_i$. After combination at the output of the FPBS, the photons exhibit an entangled state of $|H\rangle_s |H\rangle_i + e^{i\theta} |V\rangle_s |V\rangle_i$, where θ is an unknown but constant relative phase between the H and V components of the pump, which has been mentioned previously. The default signal and idler bands in our system are defined at 1550.12 nm (ITU channel 34) and 1558.17 nm (ITU channel 24), respectively. These bands were chosen because they are close enough to the pump to help inhibit Raman generation [10], but not too close to experience pump leakage. We also experimented with other bands for multichannel usage, as will be reported. In such a case, more add–drop multiplexers are concatenated to drop the desired wavelengths. The DWDM filters used have $\sim$1 nm bandwidth, and they each provide $>$60 dB suppression at the pump wavelength. The individual bands are further filtered to ensure insignificant amount of pump leakage ($>$100 dB overall pump extinction).

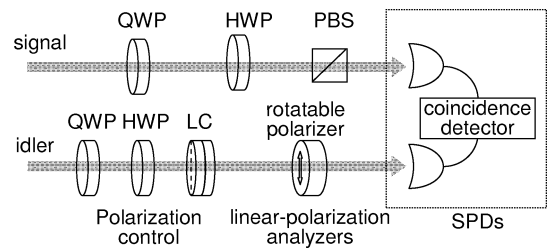

Fig. 2. Diagram of the measurement apparatus, consisting of polarization controllers, linear-polarization analyzers, and SPDs used to measure the entangled-photon pairs.

The measurement apparatus shown in Fig. 2 is composed of three sections—polarization controller, polarization analyzer, and SPD. The polarization controllers consisting of series of wave plates are used for synchronizing the polarization frame-of-reference for both the signal and idler arms prior to the polarization analyzers. The polarization analyzer at the idler arm is a rotating polarizer (hence, a linear-polarization analyzer), and the signal-arm analyzer consists of a half-wave plate (HWP) and a fixed polarization beam splitter (PBS). Essentially, the HWP and the PBS in the idler arm function as a rotating polarizer. The SPDs are based on APDs from Princeton Lightwave. The detectors were set to run with 1 ns gating window and a 2 MHz repletion rate. The dc bias, gate magnitude, and SPD temperature were adjusted to keep after-pulsing below 1%, dark count probability at $\sim$0.1%, and detection efficiency near 30%. The photon counts were counted and correlated in a custom circuit, which recorded the statistics every 1 million gates.

The polarization analyzer alignment scheme employed takes advantage of the fact that the alignment source has the same polarization state as the pump light at the input and output of the FPBS. The balanced powers of the counterpropagating lights in the Sagnac loop allow the polarization state of the alignment source to exit the FPBS in the same polarization as that of the pump source—$\mathbf{E}_{\mathrm{H}} + e^{i\theta}\mathbf{E}_{\mathrm{V}}$, where $|\mathbf{E}_{\mathrm{H}}| \approx |\mathbf{E}_{\mathrm{V}}|$ and θ is the pump phase mentioned earlier.

To be able to perform TPI measurement at any arbitrary basis, a signal photon and an idler photon with the same arbitrary polarization state at the output of the FPBS must also exhibit

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the same polarization state at the input to their respective polarization analyzers. Since they travel through two separate fibers, they experience two different Jones matrix transformations, each having three degrees of freedom (DOF) if the common phase terms are ignored and assuming isotropic media. The goal is to control the polarization in each arm independently so that the photons from both signal and idler bands would experience the same polarization transformations from the output of the FPBS to their respective polarization analyzers.

Here, we will quickly explain the physical alignment procedure followed by a more detailed conceptual explanation. The axes of the signal-arm PBS, the idler-arm polarizer, and a liquid crystal (LC) variable-phase retarder placed before the idler polarizer are all aligned to the horizontal position with respect to the table, which defines the $|H_{\mathrm{meas}}\rangle$ state of the measurement frame of reference. The signal and idler HWPs and quarter-wave plates (QWPs) are adjusted to maximize the singles counts from the alignment source. At this point, the horizontal and vertical (H_{meas} and V_{meas}) bases have been defined at the polarization analyzers, and a TPI fringe can be taken with respect to H_{meas} basis by turning OFF the alignment source and recording coincidence counts as a function of the angle of the rotatable polarizer. To complete the alignment procedure to be able to perform TPI measurements in other bases, the signal HWP is rotated 22.5° and the idler polarizer is rotated 45°, thereby effectively setting both linear polarization analyzers to the diagonal state D_{meas}. The LC control voltage is then tuned until the coincident count rate is maximized. At this point, the polarization reference frame of both the signal and idler analyzers are aligned and TPI fringes can be taken in any linear basis.

The alignment procedure is based on observing that once the singles counts from the alignment source has been maximized, the classical alignment source transforms from a polarization state at the output of the FPBS $\mathbf{E}_{\mathrm{H}} + e^{i\theta}\mathbf{E}_{\mathrm{V}}$ into a horizontal polarization $\mathbf{E}_{\mathrm{Hmeas}}$ at the input of the analyzers. This is analogous in a quantum sense that a photon at the FPBS output in the state $\left(|H\rangle + e^{i\theta}\,|V\rangle\right)/\sqrt{2}$ is transformed to $|H_{\mathrm{meas}}\rangle$. Similarly, the orthogonal output state would be transformed to $|V_{\mathrm{meas}}\rangle$. By setting the alignment source polarization to H_{meas} for both the signal and the idler arms, we have fixed two of the three DOFs of the Jones matrices, and TPI fringe measurements can be performed in the H_{meas} and the V_{meas} bases. The last DOF is governed by the phase between the H_{meas} and V_{meas}, and it can be tuned using the LC-phase retarder, with the LC-optic axis aligned to H_{meas}. By rotating the signal HWP by 22.5°, the signal-arm analyzer passes the diagonal polarization at the analyzer D_{meas}, where $|D_{\mathrm{meas}}\rangle_s = (|H_{\mathrm{meas}}\rangle_s + |V_{\mathrm{meas}}\rangle_s)/\sqrt{2}$. Now, an idler photon, which started out with the same polarization state as the signal photon at the output of the FPBS, would end up on a state $|\psi_{\mathrm{meas}}\rangle_i = \left(|H_{\mathrm{meas}}\rangle_i + e^{i\phi}\,|V_{\mathrm{meas}}\rangle_i\right)/\sqrt{2}$. When $\phi = 0$, $|\psi_{\mathrm{meas}}\rangle_i = |D_{\mathrm{meas}}\rangle_i$. Thus, by zeroing ϕ using the LC retarder, we can obtain maximum coincidence counts, which happen at $|D_{\mathrm{meas}}\rangle_i\,|D_{\mathrm{meas}}\rangle_s$. In this paper, only linear polarization analyzers are used because our measurements are all performed in the linearly polarized states—H_{meas}, V_{meas}, D_{meas}, and A_{meas}, where A_{meas} represents the antidiagonal state. A full polarization analyzer can be constructed using a rotating polarizer preceded by a QWP.

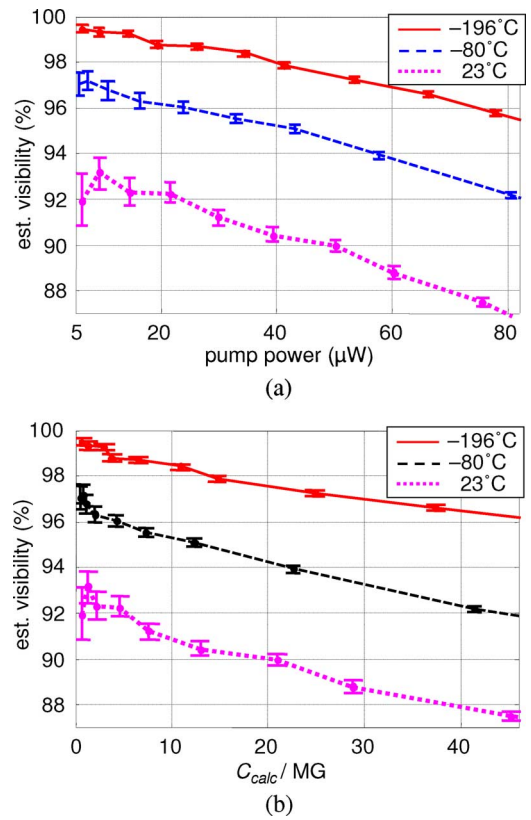

Fig. 3. (a) Plot of the estimated TPI visibility as a function of pump power coupled into one end of the DSF, taken at three different temperatures. (b) Corresponding estimated visibility as a function of dark-count-subtracted coincidence rate per million detector gates.

C. Experiment

We took estimated TPI visibility at three different DSF temperature settings—liquid nitrogen (−196 °C), a scientific freezer (−80 °C), and room temperature (23 °C). The purpose of cooling the DSF below room temperature is to reduce accidental coincidence rate by reducing Raman emission [6], [9]. Fig. 3 shows the estimated visibility versus pump power, and also as a function of the calculated coincidence count rate per million detector gates. The estimated visibility is calculated using $C_{\max}$ and A_{calc}. The error bars in the figures are calculated using propagation of uncertainty $\Delta\mathrm{Vis}_{\mathrm{est}} = \sqrt{(2A_{\mathrm{calc}}\,\Delta C_{\max})^2 + (2C_{\max}\,\Delta A_{\mathrm{calc}})^2}/(A_{\mathrm{calc}} + \Delta C_{\max})^2$, where $\Delta C_{\max} = \sqrt{C_{\max}}$ and $\Delta A_{\mathrm{calc}} = \sqrt{A_{\mathrm{calc}}}$. Of course, cooling the DSF to lower temperatures offers the best performance, but when liquid nitrogen is not a practical option, a low-temperature scientific freezer can be a convenient way to boost the performance. The freezer plugs into a standard outlet and requires no special maintenance. Each coincidence count measurement is taken over 2.5 min or 300 million gates. We typically quote the coincidence counts per million gates, as opposed to a coincidence count rate per second as the latter

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is sensitive to the speed of the detectors. At very low pump powers, the visibility at the three temperatures should be able to reach 93%, 97%, and 99%. However, at these extremely low coincidence count rates, full TPI measurements may be too noisy to be meaningful, so we tend to work at a level where peak coincidence count rate is no less than 5 counts per million gates. It is worth noting that at room and freezer temperatures, the visibility peak limitation by Raman emission can be observed at the extremely low pump-power range, where the TPI visibility actually begins to fall as the pump power is reduced. The pump powers used were not low enough to see such a peak for liquid nitrogen temperature measurement.

Fig. 4(a) and (b) shows full TPI fringe measurements with the signal arm set to four different reference basis positions—H_{meas}, D_{meas}, V_{meas}, and A_{meas}. The fringe is taken by recording coincidence counts while rotating the idler arm polarizer. Each coincidence count measurement is taken for 50 s or 10^8 gates. At room temperature, a 30 μW pump power (defined as the pump entering one end of the DSF) is used, which, according to Fig. 3, should yield >90% visibility and eight peak coincidence counts per million gates. We find good agreement with these predictions when using a sine-fitting program based on the Trust–Region algorithm [17] in MATLAB to optimize four parameters (a, b, c, d) of the equation $y = a\sin(bx + c) + d$. We note that the effect of pump-power drift is visible in Fig. 4(a) and (b) as a variation of the maximum coincidence counts from run to run.

With liquid nitrogen cooling, 45 μW pump power is used, which leads to a predicted visibility above 97% and a peak coincidence count rate of roughly 17 per million gates. As shown in Fig. 4(b), all four bases measurements match expectations reasonably well. The singles counts per million gates of both idler and signal for the four different bases measurements are shown in Fig. 4(c). As expected, there is no apparent polarization dependence of the signal or idler singles.

Due to the broad phase-matching bandwidth in the fiber, multiple signal- and idler-channel pairs can be generated using a single entanglement source. However, as the signal and idler become spaced further apart, Raman scattering can reduce performance. Thus, for a given desired performance level, there is a limited bandwidth over which entangled pairs can effectively be used. We investigate this effect by simultaneously filtering three different photon pairs. We concatenate three stages of paired channels on the ITU grid using add–drop multiplexers, as shown in Fig. 5(a). Stage 1 consists of channels 23 and 35 (1558.98 and 1549.32 nm), stage 2 consists of channels 24 and 34 (1558.17 and 1550.12 nm), and the last stage consists of channels 25 and 33 (1557.36 and 1550.92 nm). We first drop the entangled-photon pairs that are furthest in wavelength from the pump, so that they see the smallest total insertion loss, as this helps to even out the performance of the three channel pairs. Although not employed in our experiments, the use of a single WDM demultiplexer instead of concatenated ADMs would make the insertion loss of all the channels more even, and thus would increase the relative coincidence count rates of the entangled pairs closer in wavelength to the pump. It is likely that in a network distribution system, the channels with the highest two-photon generation rates would be used to traverse the longest link distances.

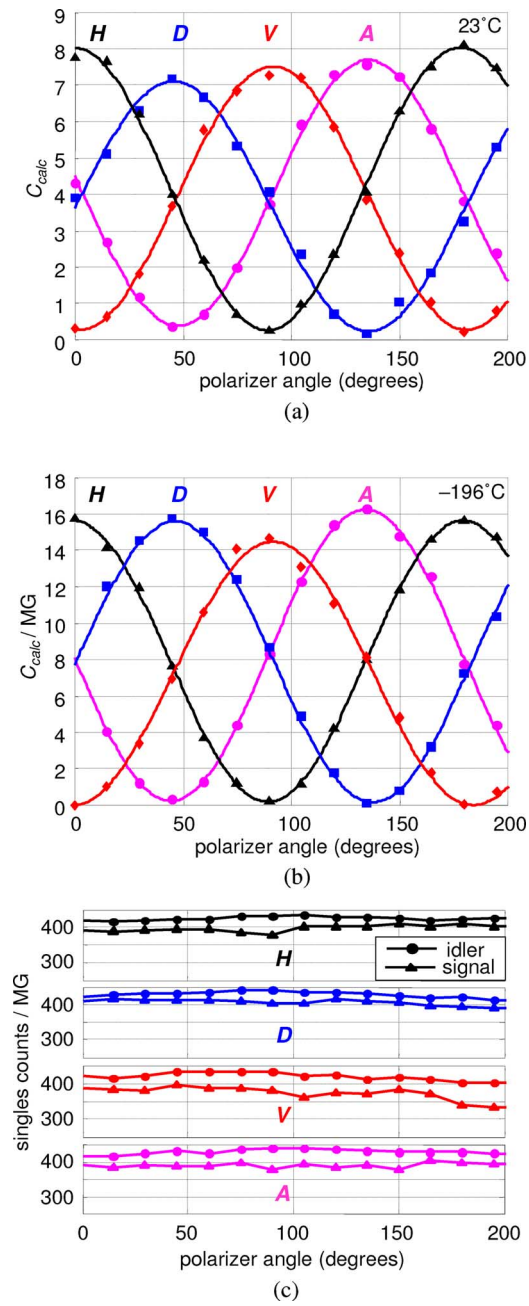

Fig. 4. (a) Measurements of coincidence counts per million gates versus rotatable polarizer angle with the signal-arm analyzer set to the H, D, V, and A positions. Data are taken at room temperature. (b) Measurements of coincidence counts versus polarizer angle taken, while the DSF is cooled in liquid nitrogen. (c) Singles counts per million gates corresponding to data in (b).

The data for each channel are taken by sending the desired channel into the polarization analyzer and aligning the analyzer accordingly. No changes to the source are made, thus demonstrating the generation of entangled photons in multiple band pairs.

The estimated visibility versus pump power and coincidence count rates at liquid nitrogen temperature are plotted in Fig. 5(a) and (b). The extra loss experienced in the second and third

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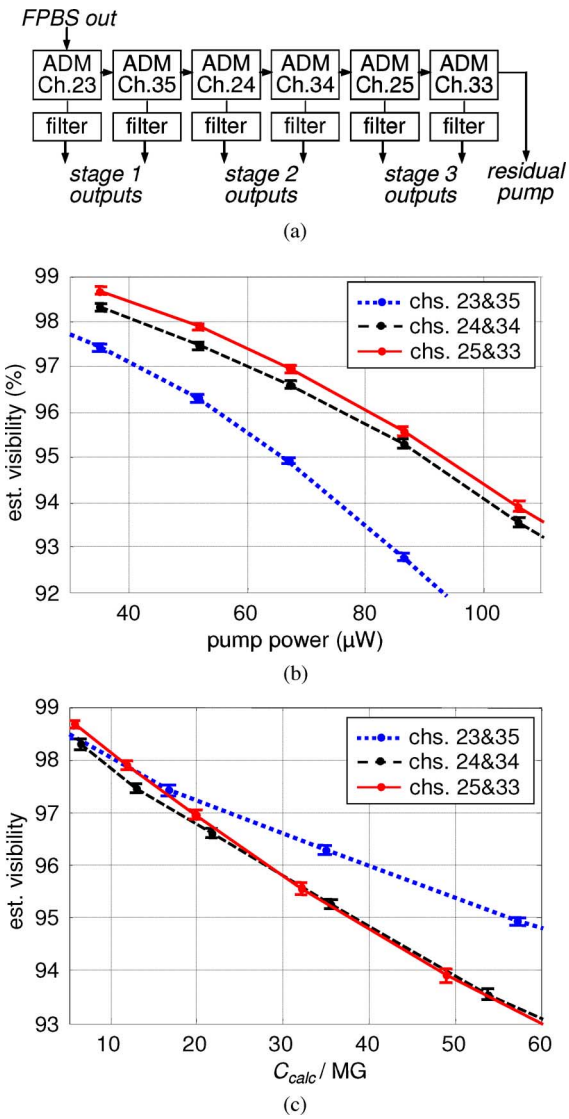

Fig. 5. (a) Configuration for filtering of three entangled-photon channel pairs. (b) Estimated TPI visibility as a function of coincidence rate per million gates for each of the three channel pairs. (c) Corresponding estimated visibility as a function of coincidence rate per million gates.

stages does not affect the estimated visibility of these channels; however, the coincidence count rates are reduced. The pump power for the group TPI measurement was chosen at 65 μW to ensure that all channels would have a visibility of roughly 95% and above, as well as a peak coincidence rate of 20 per million gates or more. Fig. 6 shows the TPI of all three channels, and from the fitted sine curve, the visibilities from the first to the last stages were 95%, 98%, and 97%, respectively. The visibility of the signal–idler pairs spaced furthest from the pump is clearly degraded, as expected from the estimated visibility plot. The slightly higher than expected visibility of the middle channel is likely due to measurement noise.

To characterize the stability of the system, we monitored the stability of the visibility by first making a measurement of the maximum coincidence count value $C_{\max}$, and then monitoring the minimum coincidence count value $C_{\min}$ for an ex-

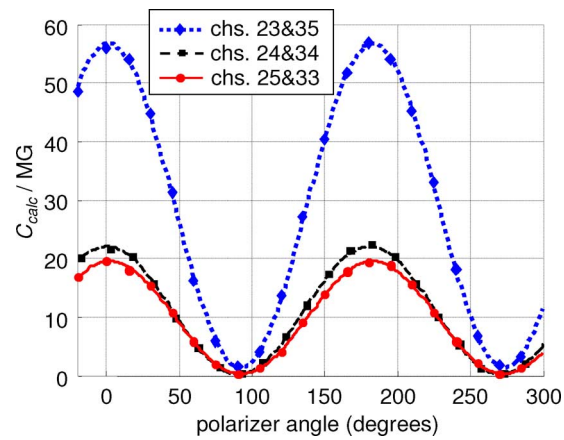

Fig. 6. TPI plot for three different channel pairs taken at pump power of 65 μW, and the visibilities of the fitted sine curves are (from top to bottom): 95%, 98%, and 97%.

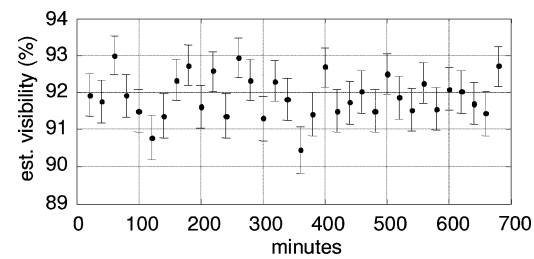

Fig. 7. Plot of the estimated visibility based on a long-term fringe-minimum measurement at room temperature, with a pump power set to 25 μW.

tended amount of time. Here, estimated visibility is calculated by $\mathrm{Vis}_{\mathrm{est}} = (C_{\max} - C_{\min})/(C_{\max} + C_{\min})$. Fig. 7 shows the results of the 12-h test by showing the visibility and assuming that the maximum coincidence is constant. The pump-power variation over the course of this test was only about $\pm 2\%$, and did not tend to drift in one direction or the other. No parts of the source or the measurement apparatus were adjusted over the course of the test. The stability of the polarization transformation through the source is thus very stable.

III. Conclusion

We describe the construction of a stable, manufacturable, multiband, polarization-entangled source using solid-core fiber. The entangled source includes a built-in alignment source, which enables a systematic method of aligning the subsequent measurement apparatus. By generating multiple WDM-entangled pairs directly in fiber at the telecommunications band, the source is shown to be well suited for entanglement distribution in a quantum-network environment.

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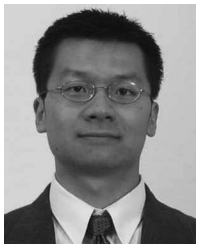

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